

Week 07: Monday, AST 5011: Astrophysical Systems

The Sun, Helioseismology, and Neutrinos

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With contributions totally ripped off from Carl Fields (UA), Mike Zingale (SUNY), Cole Miller (UMD), and Abi Nolan (Purdue).

The Present-Day Sun

Composition

The Sun is the only star we can study in extraordinary detail. Its proximity allows us to measure neutrino fluxes, perform detailed helioseismology, and resolve surface features like sunspots.

Most values for the solar composition are from Asplund et al. (2009) — "A09" in MESA.



metallicity

The mass fractions of hydrogen (X), helium (Y), and metals (Z) for widely-used compilations of the solar chemical composition. Credit: Asplund et al. 2009.

The present-day photospheric values are approximately:

- $X \approx 0.7381$ (hydrogen)
- $Y \approx 0.2485$ (helium)
- $Z \approx 0.0134$ (metals)



solar_abundance

The present-day solar photospheric elemental abundances as a function of atomic number, on the logarithmic abundance scale:

$$\log \epsilon_H = 12.00, \quad \log \epsilon_X = \log(N_X/N_H) + 12.00$$

The solar abundance problem: the 2005 downward revision of C, N, O abundances changed the opacity profile, which changed the predicted convection zone depth and worsened agreement with helioseismology. This tension remains partially unresolved.

Solar Structure: Characteristic Frequencies

The local adiabatic sound speed is:

$$c_s^2 = \Gamma_1 \frac{P}{\rho}$$

For an ideal gas: $c_s = \sqrt{\Gamma_1 k_B T / (\mu m_p)}$.

The Brunt-Väisälä frequency N describes buoyancy oscillations:

$$N^2 = g \left[\frac{1}{\Gamma_1 P} \frac{dP}{dr} - \frac{1}{\rho} \frac{d\rho}{dr} \right]$$

In convective zones ($\nabla > \nabla_{\text{ad}}$), $N^2 < 0$ — fluid parcels are unstable to convection rather than oscillating.

The Lamb frequency for angular degree ℓ :

$$S_\ell^2 = \frac{\ell(\ell+1)}{r^2} c_s^2$$

These define two classes of oscillation modes:

- p-modes (pressure modes): $\omega^2 > S_\ell^2$ and $\omega^2 > N^2$. Driven by pressure; probe the sound speed.
- g-modes (gravity modes): $\omega^2 < N^2$ and $\omega^2 < S_\ell^2$. Driven by buoyancy; probe the deep interior. Not yet definitively detected in the Sun.

Helioseismology

Observation of p-modes allows us to infer the sound speed throughout the Sun. Thousands of modes have been measured with exquisite precision.



Differences between the helioseismic and predicted sound speeds as a function of depth (Serenelli et al. 2009; credit: Asplund et al. 2009).

The largest discrepancies are at the base of the convection zone — the tachocline region, where the Sun transitions from differential rotation (convective envelope) to nearly rigid rotation (radiative interior). Recall from Lecture 10 that we saw the tachocline in the context of rotation and the Eddington-Sweet circulation.

Helioseismology constrains the solar model so tightly that when the predicted neutrino flux disagreed with observations, the fault had to lie with the neutrinos, not the solar model. This insight

led to a Nobel Prize.

Solar Luminosity Evolution

How has the Sun's luminosity changed over its 4.5 Gyr main-sequence lifetime?

Starting from the virial theorem for an ideal gas and the radiative transfer equation with Kramers opacity, we can show that:

$$L \propto \frac{M^{5.33} \rho^{0.117} \mu^{7.5}}{\kappa_0}$$

The Sun's mass changes negligibly (solar wind $\dot{M} \approx 10^{-14} M_{\odot} \text{ yr}^{-1}$). The density exponent is tiny. The Kramers prefactor κ_0 is dominated by metals and roughly constant. So the dominant factor is:

$$\frac{L(t)}{L(0)} \approx \left[\frac{\mu(t)}{\mu(0)} \right]^{7.5}$$

As hydrogen burns to helium, the mean molecular weight $\mu = 4/(3 + 5X)$ increases, and the luminosity rises.

Combining with $dX/dt = -L/(MQ)$ where $Q = 6 \times 10^{18} \text{ erg g}^{-1}$, the analytic solution is:

$$L(t) = L(0) \left[1 - \frac{85}{8} \frac{\mu(0) L(0)}{MQ} t \right]^{-15/17}$$

Fitting $L(t_{\odot}) = L_{\odot}$ at $t_{\odot} = 4.5 \times 10^9 \text{ yr}$ with $\mu(0) \approx 0.6$ gives $L(0) \approx 0.79 L_{\odot}$.

This is the faint young Sun paradox: the Sun was ~20-30% dimmer when the Earth formed, yet geological evidence shows liquid water existed. The leading explanation is a stronger greenhouse effect (higher CO_2 , CH_4) in the early atmosphere.

Exercise 1: Solar Luminosity Evolution

The Sun's luminosity evolves as:

$$L(t) = L(0) \left[1 - \frac{85}{8} \frac{\mu(0) L(0)}{MQ} t \right]^{-15/17}$$

where $\mu(0) = 0.6$ and $Q = 6 \times 10^{18} \text{ erg g}^{-1}$.

Tasks:

1. Find $L(0)/L_{\odot}$ by requiring $L(t_{\odot}) = L_{\odot}$ at $t_{\odot} = 4.5 \times 10^9 \text{ yr}$. Solve numerically with `brentq`.

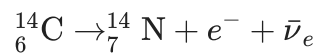
2. Plot $L(t)/L_{\odot}$ from $t = 0$ to $t = 10$ Gyr.
3. At what age does the Sun reach $2 L_{\odot}$?
4. How much dimmer was the Sun when Earth formed?

► Solution

Solar Neutrinos

History

In 1930, Wolfgang Pauli proposed a new particle to save energy and momentum conservation in beta decay:



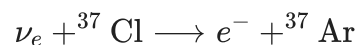
Without the neutrino, the energies and momenta of the nitrogen and electron didn't add up. Pauli originally called it the "neutron"; after Chadwick discovered the actual neutron, Pauli's particle was renamed "neutrino" (Italian for "little neutral one") following a joke by Edoardo Amaldi, popularized by Enrico Fermi. Fermi's paper on neutrinos was rejected by *Nature* as "too remote from reality."

The neutrino was first detected in a laboratory in 1956, and natural neutrinos were first detected in 1965 in a mine in South Africa.

The Solar Neutrino Problem

Starting in the 1960s, Ray Davis used a detector in the Homestake gold mine in Lead, South Dakota to detect neutrinos from the Sun. The theorist on the project was John Bahcall.

Detection method: 600 tons of cleaning fluid containing chlorine:



Threshold: ≥ 0.81 MeV. Since pp-chain neutrinos only reach 0.42 MeV (>90% of solar neutrinos), the experiment was primarily sensitive to ${}^8\text{B}$ and CNO neutrinos.

Result: 2.56 ± 0.23 SNU, where 1 SNU ("snoo") = 10^{-36} captures s^{-1} target $^{-1}$.

Predicted: 7.6 ± 1.2 SNU.

Only 1/3 of the expected neutrinos were detected! Confirmed by many subsequent experiments.

 solar_neutrinos

Neutrino Flavor Oscillations

Three lepton flavors — electron, muon, tau — each with an associated neutrino (ν_e, ν_μ, ν_τ).

Key insight: flavor eigenstates are not mass eigenstates. An electron neutrino is a quantum superposition of three mass eigenstates with energies E_1, E_2, E_3 . Each evolves as $e^{-iE_it/\hbar}$. Since the energies differ, the phases evolve at different rates, and the flavor composition changes during propagation.

A neutrino produced as ν_e in the Sun's core arrives at Earth as a mixture of ν_e, ν_μ , and ν_τ . Homestake detected only ν_e — seeing 1/3 is exactly what oscillations predict!

The MSW effect (Mikheyev-Smirnov-Wolfenstein) enhances the conversion probability in dense matter like the Sun's core.

This requires neutrinos to have mass — but very little (sum of all three flavor masses $< 10^{-6} m_e$).

Once helioseismology tightly constrained the solar model, the neutrino deficit had to be new physics. Nobel Prize (2015, Kajita and McDonald).

Neutrino Cross Sections

Neutrino-electron scattering: $\nu_e + e^- \rightarrow \nu_e + e^-$

$$\sigma_\nu \approx 10^{-44} \left(\frac{E_\nu}{m_e c^2} \right)^2 \text{ cm}^2$$

For comparison, the Thomson cross section is $\sigma_T \approx 6.65 \times 10^{-25} \text{ cm}^2$ — about 1 "barn" (10^{-24} cm^2). A typical neutrino cross section is about 10^{-44} cm^2 , or about 10,000 "sheds" (1 shed = 10^{-48} cm^2). As particle physicists say: hitting an electron with a photon is like hitting the broad side of a barn; hitting it with a neutrino is... not.

The $\sigma \propto E^2$ scaling holds up to the rest mass of the target. Above that, $\sigma \propto E$, then levels off. High-energy neutrinos are easier to detect (but rarer).

Exercise 2: Solar Neutrino Optical Depth

The optical depth for neutrinos traversing the Sun is:

$$\tau_\nu = \bar{n}_e \sigma_\nu R_\odot$$

Tasks:

1. Estimate $\bar{n}_e \approx \bar{\rho}/(2 m_p)$ (assuming fully ionized H/He).
2. Compute σ_ν for a 1 MeV neutrino.

3. Compute τ_ν .
4. What fraction of solar neutrinos interact before escaping?

► Solution

Exercise 3: Neutrino Mean Free Path — Sun vs Supernova

The mean free path is $\lambda = 1/(n_e \sigma_\nu)$.

In the solar core: $\rho_c \approx 150 \text{ g cm}^{-3}$, $E_\nu \sim 1 \text{ MeV}$.

In a core-collapse supernova: $\rho \sim 3 \times 10^{14} \text{ g cm}^{-3}$, $E_\nu \sim 10 \text{ MeV}$.

Tasks:

1. Compute the neutrino mean free path in the solar core.
2. Compute the neutrino mean free path in a SN core.
3. Compare each to the relevant physical size ($R_\odot$ for the Sun, $R_{\text{NS}} \approx 10 \text{ km}$ for the proto-neutron star).
4. In which environment are neutrinos trapped?

► Solution

SN 1987A and Modern Neutrino Detectors

SN 1987A (February 24, 1987, Large Magellanic Cloud): approximately 20 neutrinos with energies of tens of MeV were detected across a few detectors worldwide. This confirmed the basic picture of core collapse and constrained properties of the neutrino.

Modern detectors are vastly larger. A Galactic supernova today would yield tens to hundreds of thousands of neutrinos — an unprecedented view of the core collapse process.

IceCube (South Pole): strings of photomultipliers distributed through $\sim 1 \text{ km}^3$ of Antarctic ice. When a high-energy neutrino strikes a nucleus, the recoiling particle travels faster than the speed of light *in ice* (the universal speed limit is c in vacuum, not in a medium). This produces Cherenkov radiation detected by the photomultipliers. IceCube has detected neutrinos up to PeV energies from extragalactic sources.

Solar Sound Speed and Oscillation Frequencies

For an ideal gas in the solar interior (fully ionized, $\mu \approx 0.6$, $\Gamma_1 \approx 5/3$):

$$c_s = \sqrt{\frac{\Gamma_1 k_B T}{\mu m_p}}$$

The p-mode frequencies for radial modes are approximately:

$$\nu_{n,0} \approx \Delta\nu \left(n + \frac{1}{4} \right)$$

where n is the radial order and the large frequency spacing is:

$$\Delta\nu \approx \left(2 \int_0^R \frac{dr}{c_s} \right)^{-1}$$

This is roughly the inverse of the sound crossing time. For the Sun: $\Delta\nu_{\odot} \approx 135 \mu\text{Hz}$ and $\nu_{\text{max},\odot} \approx 3090 \mu\text{Hz}$.

The acoustic cutoff frequency $\nu_{\text{ac}} = c_s / (4\pi H_P)$ at the photosphere sets the upper limit for trapped p-modes.

Exercise 4: Solar Sound Speed and Oscillation Frequencies

Tasks:

1. Compute c_s at the solar center ($T_c = 1.57 \times 10^7 \text{ K}$, $\mu = 0.6$, $\Gamma_1 = 5/3$) and at the base of the convection zone ($T \approx 2 \times 10^6 \text{ K}$).
2. Estimate the sound crossing time $t_{\text{cross}} \sim 2 R_{\odot} / \bar{c}_s$ using the geometric mean.
3. Compare $\Delta\nu \approx 1/t_{\text{cross}}$ with the observed $135 \mu\text{Hz}$.
4. Compute the acoustic cutoff $\nu_{\text{ac}} = c_s / (4\pi H_P)$ at the photosphere ($T = 5777 \text{ K}$, $\mu = 1.3$, $H_P \approx 150 \text{ km}$) and compare with $\nu_{\text{max},\odot} = 3090 \mu\text{Hz}$.

► Solution

From Helioseismology to Asteroseismology

The techniques developed for the Sun have been extended to other stars through asteroseismology. Two global seismic quantities scale with stellar properties:

$$\nu_{\text{max}} \propto \frac{M}{R^2 \sqrt{T_{\text{eff}}}}, \quad \Delta\nu \propto \sqrt{\frac{M}{R^3}}$$

Calibrated to the Sun, these scaling relations determine stellar masses and radii from photometry alone. The *Kepler* mission measured oscillations in thousands of stars, revolutionizing our knowledge of stellar interiors.

In the following in-class assignment, you will apply these techniques to real *Kepler* data.

In-Class Assignment

Credit: Oliver Hall (`oliver.hall@esa.int`), Geert Barentsen

How to estimate a star's mass and radius using asteroseismology

Learning Objectives

In this tutorial you will learn:

- What the periodogram of a star with solar-like oscillations looks like.
- How the oscillations can be characterized by two key metrics ($\nu_{\max}$ and $\Delta\nu$).
- How these metrics can be used to estimate a star's mass, radius, and surface gravity.

Introduction

Stars of all shapes and sizes oscillate. In many cases, these oscillations are created by standing waves formed inside of the star, which disturb the stellar surface. The study of these types of internal oscillations is called *asteroseismology*.

This tutorial focuses on a very common type of oscillating star: *solar-like oscillators*. This type includes main sequence stars like the Sun as well as many red giant stars. Stars of this type experience convection in the outer layers of their atmospheres. As material moves up and down in these layers, the turbulent motion forms damped standing waves throughout the stellar interior. Because these waves probe the full stellar interior, we can estimate fundamental properties of a star by studying how these waves disturb the surface of the star. Properties which can be inferred in this way include mass, radius, surface gravity, and even age for the most advanced studies.

In this tutorial we will explain two key metrics asteroseismologists use to characterize the oscillations of a Sun-like star: the average frequency spacing $\Delta\nu$, and the frequency of maximum oscillation amplitude $\nu_{\max}$. We will also demonstrate how these metrics can be used to estimate the mass, radius, and surface gravity of a star.

Imports

This tutorial requires **Lightkurve** (for its features to handle *Kepler* data) and **NumPy** (to help plot some additional details on figures). We will use **Matplotlib** for plotting.

a - Query and download the data for the solar-like oscillator KIC 10963065

1. Search for and download and stitch the data using `lk.search_lightcurve` and the following arguments:

- `target = 'KIC 10963065'`
- `cadence='short'`,
- `author='Kepler'`,
- `quarter=(2,5,6,7)`

You can use the `lc.meta` command to output metadata of the lightcurve including the time units.

The frequency spectrum (also known as periodogram) of a solar-like oscillator like the Sun shows a rich spectrum of modes of oscillation, equally spaced in frequency, colloquially referred to as a "comb."

We will investigate this spectrum using the Sun-like star KIC 10963065 as an example. This star is also colloquially known as "Rudy" (cf. [Lund et al. 2016](#)). In order to get a good resolution in the frequency domain, we start by downloading and combining four quarters of Short Cadence *Kepler* data as follows:

b - Lightcurve periodogram

In one of the companion tutorials, we already identified the location of the modes of oscillation, so we'll zoom in on that region right away when creating the periodogram. We also set the `normalization='psd'` keyword argument, to match the common conventions in asteroseismology of solar-like oscillators.

1. Using Lightkurve, convert the time-series data to a periodogram (name it `pg`) showing the oscillation frequencies.
2. Uncomment the cell below to plot additional envelope plots as well.

Set the `normalization='psd'` and minimum to maximum frequency to 1500 to 2700 μHz .

In this spectrum we can see that the modes of oscillation rise up from the noise, reach a peak, and then decrease in power. This region, where the modes are visible, is commonly referred to as the *mode envelope*. The peak of this envelope is what we call the *frequency of maximum oscillation amplitude*, or ν_{max} (colloquially called "numax").

An approximation of the shape of the envelope (roughly a Gaussian function) is shown as the dashed line in the figure below. We also overlay a smoothed version of the power spectrum, and annotate

the location of the oscillation modes with arrows. Making this figure is an optional step, but it helps to visualize the oscillation features:

A distinctive feature in the spectrum is the equal spacing of the modes of oscillation. The spacing between two consecutive overtones of the same radial order (for example, radial oscillations, dipole oscillations, etc.) is what we call the *average large frequency spacing*, or $\Delta\nu$ (colloquially called "deltanu"). In the figure above, the arrows point to dipole modes of oscillation (hemispheres expanding and contracting alternately), and are all evenly spaced.

Both of these metrics, $\nu_{\max}$ and $\Delta\nu$, are very useful because they are related to the mass, radius, and temperature of the star. The next step in our tutorial is to estimate the values of these metrics, using Lightkurve's tools.

c. - Estimating the Frequency of Maximum Amplitude $\nu_{\max}$

The frequency of maximum amplitude can be estimated in a number of ways. An obvious estimate would be to take the frequency that corresponds to the maximum amplitude in the power spectrum. This would not be a robust estimate however, because the maximum peak is very sensitive to noise and sampling effects.

Instead, one of the most commonly used methods to estimate $\nu_{\max}$ is the 2D autocorrelation function (ACF) method. Literature publications using this method include, for example, [Huber et al. \(2009\)](#) and [Viani et al. \(2019\)](#). A version of this method is provided by the Lightkurve package.

Before diving into Lightkurve's tools, we will need to prepare our [Periodogram](#) object.

1. First, remove background noise using the [flatten\(\)](#) method and plot the result.
2. Next, pass this flatten version of our data to construct our `Seismology` object.
3. Plot an overview of our Seismology object using: `seismology.diagnose_numax()`;

Lightkurve's estimation method uses the autocorrelation function (ACF). Autocorrelation values are the result of calculating the correlation of the data with itself as it is gradually shifted over itself. One way to imagine how the ACF method works is by holding your hands up side by side and slowly shifting them past one another. Where your fingers overlap, the ACF will be high, and vice versa.

Deeper look at the output of `diagnose_numax`

The figure above shows three plots, so let's break them down, one by one:

1. The top panel shows the original periodogram.
2. The middle panel shows an image representing the ACF values calculated for different parts of the power spectrum. In this case, each segment used to calculate the ACF is $250 \mu\text{Hz}$ wide,

which you can see on the y-axis. The x-axis is the central frequency of each segment, and the z-axis (the color) shows the strength of the autocorrelation function. As we get closer to the frequency of maximum oscillation, the colors get darker, and so the ACF is increasing.

3. The bottom panel shows the same autocorrelation values as the middle panel, but collapsed (that is, summed) along the y-axis. The integrated value of the ACF is strongest closest to the frequency of maximum oscillation. The blue line over the top is a smoothed version of the collapsed ACF. Lightkurve estimates $\nu_{\max}$ by computing the maximum of the smoothed blue line. The resulting estimate is marked across all three panels using the vertical red line.

Now that we have estimated a value for $\nu_{\max}$, calculating $\Delta\nu$ becomes faster. We can observe the features of $\Delta\nu$ in the middle panel above: there are repeating peaks in the ACF, corresponding to repeating features overlapping, just like the fingers on your hand. The spacing between these repeating peaks will be used to estimate $\Delta\nu$.

d - Estimating the Frequency Spacing $\Delta\nu$

1. In a similar method, use `seismology.diagnose_deltanu()` to plot our deltanu estimates.

In order to estimate $\Delta\nu$ efficiently, we can use the fact that we only need to investigate the ACF in the region surrounding $\nu_{\max}$. In this region, peaks in the ACF will show up corresponding to the repeating comb structure overlapping with itself.

Having an estimate of $\nu_{\max}$ also makes measuring $\Delta\nu$ possible in a second way. Using well-established empirical relations which have been published in the literature, we can infer the Full Width Half Maximum (FWHM) of the mode envelope based on the value of $\nu_{\max}$. This helps us further narrow down the region in which to calculate the ACF. Lightkurve uses the relationships published by [Mosser et al. 2010](#) and [Lund et al. 2017](#), which are:

$$\text{FWHM} \approx 0.25 \times \nu_{\max} \text{ for main sequence stars, and}$$

$$\text{FWHM} \approx 0.66 \times \nu_{\max}^{0.88} \text{ for red giants.}$$

We can also use $\nu_{\max}$ to make a rough estimate of $\Delta\nu$ to guide our search for the true $\Delta\nu$. This is roughly

$$\Delta\nu \approx 0.294 \times \nu_{\max}^{0.772},$$

which we assume to be the same for all solar-like oscillators ([Stello et al. 2009](#)).

To understand how this works in practice, let's first `estimate_deltanu()`, and then `diagnose_deltanu()`.

In the top panel we see the region in the frequency spectrum for which the ACF is evaluated, which is one FWHM on either side of $\nu_{\max}$. This region encompasses the strongest modes of oscillation.

In the bottom panel we can see the ACF, that is, the result of calculating the correlation of the data with itself as it is shifted over itself. Where modes of oscillation overlap with one another, there will be a spike in the ACF. Because the modes are regularly spaced, these spikes will also be regularly spaced, and will reflect the spacing between modes of oscillation.

The inset on the upper right hand corner shows a region around the *empirically* estimated $\Delta\nu$, which is based on the literature equations shown above. Using the [Scipy](#) `find_peaks()` function, Lightkurve finds the tip of the ACF peak nearest to this estimate, and reports this as the *measured* $\Delta\nu$.

e - Estimating Stellar Mass, Radius, and Surface Gravity

1. Using our Seismology object to retrieve the mass, radius, and logg on our object.

Having obtained estimates for $\Delta\nu$ and $\nu_{\max}$, we can now estimate the star's mass (M), radius (R), and surface gravity (g) given an estimate for the star's effective temperature (T_{eff}).

Asteroseismologists have established three *scaling relations* between these properties. These scaling relations read as follows:

$$\frac{M}{M_{\odot}} \simeq \left(\frac{\nu_{\max}}{\nu_{\max,\odot}} \right)^3 \left(\frac{\Delta\nu}{\Delta\nu_{\odot}} \right)^{-4} \left(\frac{T_{\text{eff}}}{T_{\text{eff},\odot}} \right)^{3/2},$$

$$\frac{R}{R_{\odot}} \simeq \left(\frac{\nu_{\max}}{\nu_{\max,\odot}} \right) \left(\frac{\Delta\nu}{\Delta\nu_{\odot}} \right)^{-2} \left(\frac{T_{\text{eff}}}{T_{\text{eff},\odot}} \right)^{1/2} \text{ and}$$

$$\frac{g}{g_{\odot}} \simeq \left(\frac{\nu_{\max}}{\nu_{\max,\odot}} \right) \left(\frac{T_{\text{eff}}}{T_{\text{eff},\odot}} \right)^{1/2}$$

where the $\odot$ symbol represents the Sun. In Lightkurve, the solar values used are $\nu_{\max,\odot} = 3090 \mu\text{Hz}$, $\Delta\nu_{\odot} = 135.1 \mu\text{Hz}$ ([Huber et al. 2010](#)) and $T_{\text{eff},\odot} = 5777.2 \text{ K}$ ([Prša et al. 2016](#)).

These scaling relations are built into Lightkurve. We can call them as follows:

5. Closing Remarks

In this tutorial, we have seen how Lightkurve's tools can be used to estimate the mass, radius, and surface gravity of a solar-like oscillator. We did this by first estimating the seismic observables $\nu_{\max}$ and $\Delta\nu$. It is important to remember, however, that these are *estimates*, and that the method built into Lightkurve is an adaptation of established 2D ACF methods designed to provide first look estimates of stellar properties. Lightkurve's implementation has not been tested thoroughly in the literature yet, and it does not provide uncertainties yet.

If you're interested in learning more about asteroseismology, we recommend the following review papers:

- [Chaplin & Miglio \(2013\)](#) provides a review on asteroseismology of solar-like oscillators with *Kepler*, including an explanation of the seismic scaling relations.
- [Aerts \(2019\)](#) provides a comprehensive review that covers asteroseismology of a wide range of oscillating stars, including solar-like oscillators.

Or even better, get in contact and collaborate with an asteroseismologist!